

Multiple Choice (10 marks)

1. The atomic mass of the ^4He atom is 4.002603 u. Find the binding energy of the ^4He nucleus in MeV.
 - (a) 33.9
 - (b) 29.8
 - (c) 27.5
 - (d) 31.2
 - (e) 24.3
2. Calculate the self-inductance of a coil if a rate of change of current of 5 amperes per second produces a backemf of 2 volts.
 - (a) 0.4 henry
 - (b) 0.3 henry
 - (c) 0.2 henry
 - (d) 0.6 henry
 - (e) 0.7 henry
3. What happens to the force of gravitational attraction between two small objects if the mass of each object is doubled and the distance between their centers is doubled?
 - (a) It is increased by two times.
 - (b) It remains the same.
 - (c) It becomes zero.
 - (d) It is reduced to half.
 - (e) It is increased by eight times.
4. What is the focal length, of a combination of two thin convex lenses, each of focal length f , placed apart at a distance of $(2f/3)$?
 - (a) $(f/3)$
 - (b) $(f/2)$
 - (c) $(5f/3)$
 - (d) $(6f/5)$
 - (e) $(3f/4)$
5. Why are the seasons in Mars' southern hemisphere so extreme?

- (a) because Mars' axis is less tilted than the Earth's
- (b) because Mars has a smaller size than the Earth
- (c) because Mars has a slower rotation speed than the Earth
- (d) because Mars has more carbon dioxide in its atmosphere than the Earth
- (e) because Mars has a shorter year than the Earth

True / False (10 marks)

Answer True or False

6. True or False: The ratio of forward-bias to reverse-bias currents at 1.5 V is -6×10^{25} , indicating a significant difference in behavior.
7. True or False: An object with an angular size of 0.5 degrees and a height of 5 cm is approximately 15 meters away from the observer.
8. True or False: A lens with a focal length of 20 in can produce a virtual image when the object is placed 10 in from it.
9. True or False: A thin film with half the optical density of a metallic coating absorbs 44% of incident light.
10. True or False: The surface temperature of the Sun is approximately 6000 K, which is consistent with stellar temperature classifications.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the change in entropy when 100 g of water at 0°C is mixed with 50 g of water at 50°C . Analyze the thermodynamic processes involved.
11. Calculate the electrostatic force acting on a 10.0 g block with a charge of $+8.00 \times 10^{-5}\text{ C}$ in the electric field $\mathbf{E} = (3000\mathbf{i} - 600\mathbf{j})\text{ N/C}$, and discuss the underlying principles involved.

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